

Parent Hamiltonian Benchmark Figs

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We construct deterministic equilibrium benchmarks for the figures of the current adaptive-circuit manuscript. The reference is the clean signed sum of normalized, hard-wall-truncated overcomplete Wannier projectors, at range $n_{\text{shell}} = 1$ and exterior parameter $\alpha_2 = 30$, with periodic boundary conditions and exact half filling. Nine figure groups reproduce the selected observables: the local Chern marker, filtered occupation spectra, the low-energy gap and mode density, correlations, strip entropy and charge variance, restricted spectra, wall entropy, modular evolution, and antipodal mutual information. A separate table gives Chern-number deviations for square systems of lengths 20, 30, 40. The clean parent closely realizes the unit entropy and charge coefficients and equal half-unit wall contributions. Its strictly periodic low-energy splitting, finite-ring correlator, and discrete spectral-window counts also expose differences from the trajectory ensemble. This note is a standalone benchmark collection; it does not modify the manuscript or its appendixes.

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I. CLEAN PARENT, GEOMETRY, AND CONVENTIONS

The overcomplete Wannier (OW) modes are generated by the canonical CPU class `classA_U1FGTN`, using its `construct_OW_projectors` method. For every center, we use the two σ^x trial orbitals and the uniform Bloch projectors at the parameter assigned to that center. We retain the square support $|d_x|, |d_y| \leq 1$, discard support on the opposite side of a hard wall, and normalize each remaining mode independently. The lattice is periodic in both directions, with no boundary twist, random mass, random projector weight, or measurement record. The central slab has parameter α_1 and the exterior has $\alpha_2 = 30$. Its inclusive coordinates are

$$x_L = N_x/2 - \lfloor N_x/4 \rfloor, \quad x_R = N_x/2 + \lfloor N_x/4 \rfloor. \quad (1)$$

Thus the walls are at (5, 15) for $N_x = 20$ and (8, 22) for $N_x = 30$. The exterior connects across the periodic x seam, but all parent matrix elements between the slab and exterior vanish.

Using the canonical stored OW columns, define the single-particle parent

$$h_{\text{par}} = \sum_{\mathbf{r}, \nu=A,B} (|W_{\mathbf{r}, \nu, +}\rangle \langle W_{\mathbf{r}, \nu, +}| - |W_{\mathbf{r}, \nu, -}\rangle \langle W_{\mathbf{r}, \nu, -}|). \quad (2)$$

This construction is often called the flattened OW parent because its ingredients come from band projectors. We retain its actual dispersive eigenvalues; we do not replace its spectrum by ± 1 . The energy normalization in Eq. (2) is fixed, with no rescaling to fit circuit data. Translation symmetry along y reduces diagonalization to N_y blocks of size $2N_x$.

There are $2N_x N_y$ physical orbitals. Filling exactly the lowest $N_x N_y$ levels gives an occupied frame V and spectral projector $P = VV^\dagger$. The canonical stored-column convention and manuscript correlation convention are related by

$$G_{ij} = \langle \hat{c}_i^\dagger \hat{c}_j \rangle = P_{ji}, \quad G = P^\top. \quad (3)$$

All Chern and modular-propagation signs use this mapping. Restricted occupations, entropy, charge variance, and orbital-summed densities are invariant under this transpose. No average over distinct states is performed. Averages over translated cuts are exact spatial averages; their spectra coincide by translation symmetry.

Numerically degenerate eigenvalues are grouped within $10^{-12} \max(1, \|h_{\text{par}}\|)$. At a partially occupied Fermi-level subspace we first retain momentum eigenstates, resolve transverse position within each momentum block, and use projected coordinate vectors to break any remaining ties. The reverse occupation choice is evaluated as a diagnostic. For the square $L = 30, 40$ cases, two Fermi-level modes are unresolved at this tolerance. Their alternative choices change the tabulated bulk Chern number by less than 10^{-12} and the half-strip entropy by less than 3×10^{-12} . At fixed $N_x = 20$, the approximately 1.50×10^{-9} occupied-to-empty splitting is resolved, so no such tie prescription is needed.

A. Relation to the manuscript and numerical statistics

Figure captions identify the corresponding current manuscript number, independently of its asset filename. Ground-state calculations replace trajectory ensembles. There is no trajectory count or sampling standard error. Joint fits give equal total weight to each size, using the manuscript's displayed coordinates and fit windows. We report deterministic residuals; a coefficient from a poor fit is not assigned a physical scaling interpretation. Logarithms are natural throughout.

II. BULK TOPOLOGY: MANUSCRIPT FIGURE 2(B,C)

The local marker is evaluated with the same ordinary periodic-coordinate seam as the manuscript:

$$C(\mathbf{r}) = 2\pi i \sum_{\mu} [GXGYG - GYGXG]_{\mathbf{r}\mu, \mathbf{r}\mu}. \quad (4)$$

Its full-system sum vanishes as a finite commutator trace; this does not set the bulk topological invariant to zero. The plot displays $\tanh C$ and the saved data also contain the untransformed marker.

For the three-sector disk estimator, use $x_0 = L/2$, $R = 0.2L$, the manuscript's counterclockwise sector boundaries, and minimum-image displacements. The estimator is

$$\mathcal{C}_G = -24\pi \operatorname{Im} \operatorname{tr}(G_{CA}G_{AB}G_{BC}). \quad (5)$$

Every integer y_0 is included. Their spread is at floating-point precision and is not a statistical uncertainty. The equilibrium table replaces the dynamical cycle-convergence panel rather than assigning fictitious cycles to a ground state.

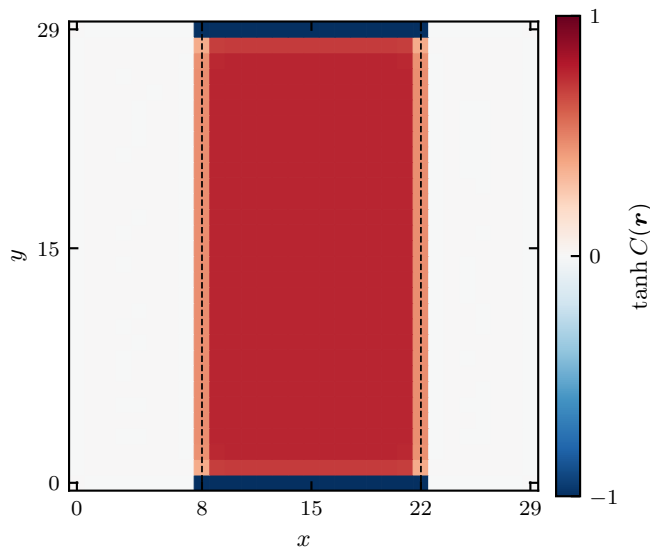


FIG. 1. Ground-state counterpart of manuscript Figure 2(b). Local Chern marker on the 30×30 torus at $\alpha_1 = 1$, with walls at $x = 8, 22$. The color scale is $\tanh C(\mathbf{r}) \in [-1, 1]$ and retains the periodic coordinate seam. There is one clean half-filled ground state and no statistical averaging.

TABLE I. Square-system ground-state counterpart of manuscript Figure 2(c). The disk centers cover the full circumference.

L	(x_L, x_R)	R	$\mathcal{C}_G$	$\mathcal{C}_G - 1$	$ \mathcal{C}_G - 1 $
20	(5, 15)	4	0.999957845235	-4.215×10^{-5}	4.215×10^{-5}
30	(8, 22)	6	0.999999926439	-7.356×10^{-8}	7.356×10^{-8}
40	(10, 30)	8	0.999999999743	-2.569×10^{-10}	2.569×10^{-10}

III. FILTERED OCCUPATIONS: MANUSCRIPT FIGURE 3(B,C)

Starting from maximal mixing, imaginary-time filtering with the many-body parent $\hat{H}_{\text{par}} = \sum_{ij} (h_{\text{par}})_{ij} \hat{c}_i^\dagger \hat{c}_j$ gives

$$\hat{\rho}(t) = \frac{e^{-2t\hat{H}_{\text{par}}}}{\text{Tr} e^{-2t\hat{H}_{\text{par}}}}, \quad n_j(t) = \frac{1}{1 + e^{2tE_j}}, \quad G(t) = [\mathbf{1} + e^{2t\mathbf{h}_{\text{par}}}]^{-\text{T}}. \quad (6)$$

This uses the same factor of two as manuscript Eqs. (13)–(14). We set t to the numerical values 1, 5, 60 of its cycle labels, with no adjustable scale. Here t is an inverse-energy filtering parameter, not a physical circuit cycle. The Gaussian state is grand canonical; exact particle-hole pairing fixes its mean filling to one half. At exactly zero energy, Eq. (6) leaves occupation 1/2 even as $t \rightarrow \infty$, whereas the separately chosen half-filled ground state is a projector.

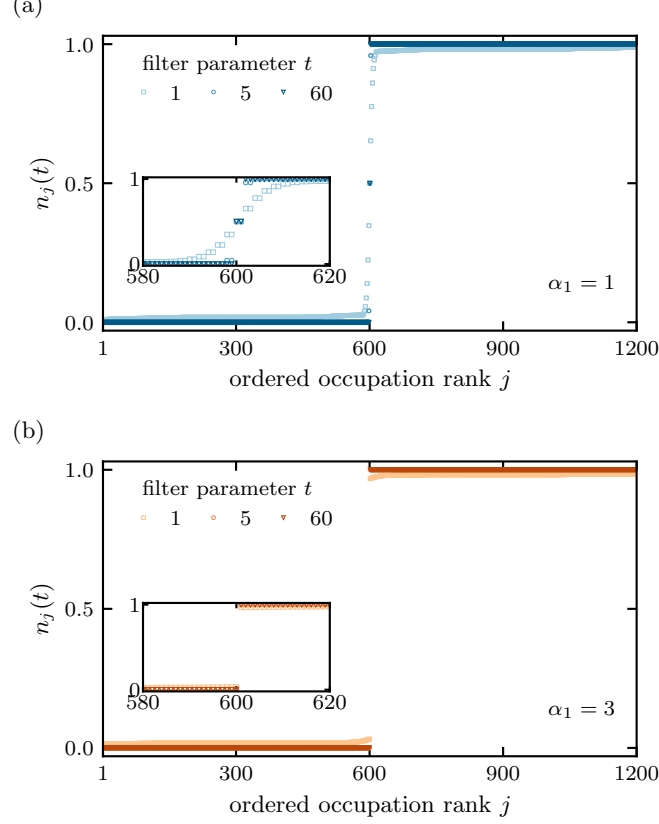


FIG. 2. Counterparts of manuscript Figure 3(b,c), shown here as panels (a,b). Ascending filtered occupation spectra at $N_x = 20, N_y = 30$ for (a) $\alpha_1 = 1$ and (b) $\alpha_1 = 3$, at $t = 1, 5, 60$. All 1200 ranks are shown; insets use identical limits on ranks 580–620. Eigenvalue ranks are reordered after applying Eq. (6). The topological parent retains two occupations near 1/2 over these times because its periodic edge splitting is extremely small. The trivial parent rapidly approaches a projector.

IV. LOW-ENERGY GAP AND DENSITY: MANUSCRIPT FIGURE 4

Define two distinct spectral gaps:

$$\delta_{\text{par}} = \min_j |E_j|, \quad g_{\text{hf}} = E_{N_x N_y + 1} - E_{N_x N_y}. \quad (7)$$

The spectrum is paired, so $g_{\text{hf}} \simeq 2\delta_{\text{par}}$. At fixed $N_x = 20$, every circumference in the plotted sweep samples $k_y = 0$. The small interwall hybridization at that momentum controls the minimum gap; increasing the circumference does not remove it. Thus this clean, periodic minimum does not reproduce the manuscript's finite-time trajectory-gap N_y^{-1} trend. We retain the measured values rather than fit them to that law. This is a boundary-sector and observable distinction, not a failure to construct the clean topological parent.

The normalized low-energy density is the projector density of all states with minimum $|E|$, divided by that subspace's dimension. It includes the positive and negative partners and is independent of their internal basis choice.

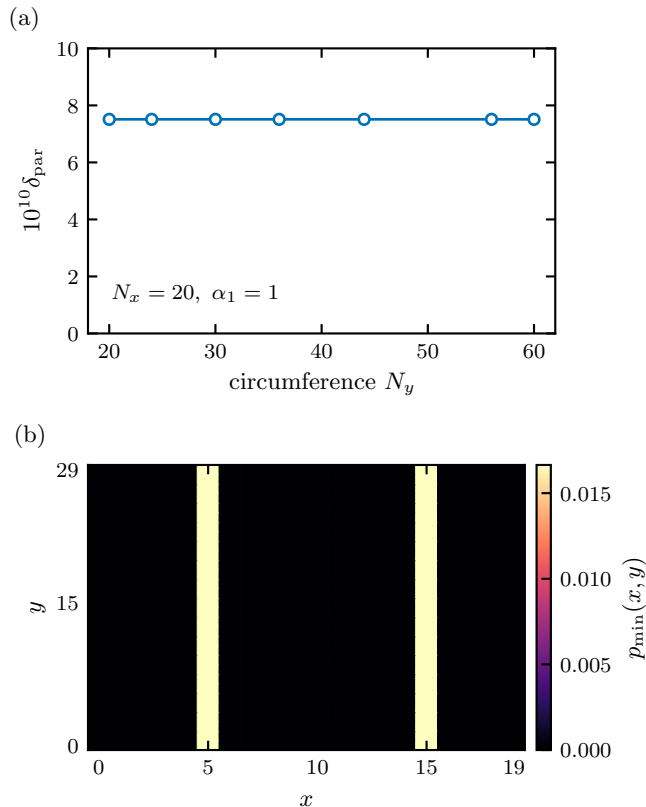


FIG. 3. Counterpart of manuscript Figure 4. (a) Minimum absolute parent energy at $\alpha_1 = 1$, fixed $N_x = 20$, and $N_y = 20, 24, 30, 36, 44, 56, 60$. The gap remains near 7.5111×10^{-10} ; connecting lines are guides. The saved CSV gives both gap definitions and the numerical resolution. (b) Normalized density of the two minimum- $|E|$ modes at $N_y = 30$. It is uniform along y and concentrated on the two walls.

V. SPATIAL CORRELATIONS: MANUSCRIPT FIGURE 5

We retain the manuscript's squared-correlator definition,

$$C_G(r_y) = \frac{1}{2N_x N_y} \sum_{x,y,\mu,\mu'} |G_{(x,y,\mu),(x,y+r_y,\mu')}|^2, \quad D(\ell) = \frac{N_y}{\pi} \sin \frac{\pi \ell}{N_y}. \quad (8)$$

The topological state has much longer-ranged correlations than the trivial state. However, the periodic clean-parent finite-ring form is not the same as the trajectory ensemble's antipodally normalized chord power. The same clean 20×60 state was studied in the repository's earlier wall-projected equilibrium analysis: the wall amplitude fits $|\cot(\pi r_y/N_y)|^\eta$ with $\eta = 1.0004103$ over $5 \leq r_y \leq 25$. The squared wall correlator therefore has exponent $2\eta \simeq 2$ in that coordinate. The common-input comparison reproduces the earlier result and is saved with the source and data hashes.

Near the antipode, the leading cotangent contribution is suppressed. Normalizing by that small residual amplifies the finite-size differences. Keeping the requested chord fit gives $\beta = 22.7291$ with weighted $R^2 = -0.9234$; this formal fit is poor and is not evidence for a large anomalous exponent. The denominator is used as computed, without a floor.

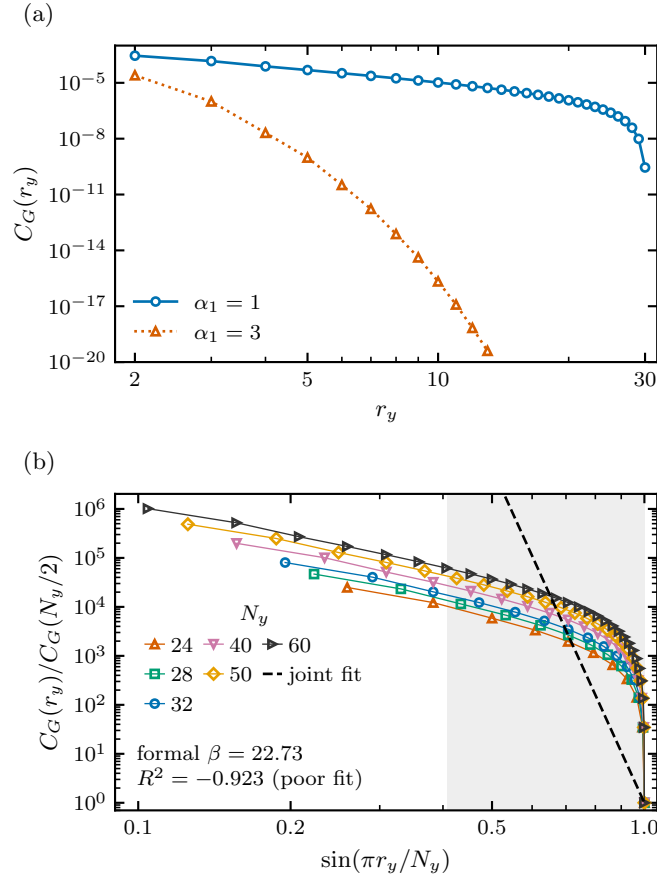


FIG. 4. Counterpart of manuscript Figure 5. (a) Ground-state squared correlations at $N_y = 60$, comparing $\alpha_1 = 1, 3$, with $N_x = 20$. Values at or below 10^{-20} are omitted from this logarithmic display only. (b) Antipodally normalized $\alpha_1 = 1$ curves for $N_y = 24, 28, 32, 40, 50, 60$ versus $\sin(\pi r_y/N_y)$. Displayed separations satisfy $r_y \geq 2$; gray shading marks the union of the $8 \leq r_y \leq N_y/2$ fitting windows. The dashed curve is the requested zero-intercept joint log fit, whose poor residual is explicitly reported. Its disagreement with the data is part of the benchmark.

VI. STRIP ENTROPY AND CHARGE: MANUSCRIPT FIGURE 6

For a full-width strip of A_y consecutive rows, let ν_a be the eigenvalues of G_A . We compute

$$S(A_y) = \sum_a h(\nu_a), \quad F(A_y) = \sum_a \nu_a(1 - \nu_a), \quad h(\nu) = -\nu \log \nu - (1 - \nu) \log(1 - \nu). \quad (9)$$

The exact endpoint limit $0 \log 0 = 0$ is used; only roundoff outside $[0, 1]$ is clipped. Subtract the reference at $A_y^* = \lfloor N_y/2 \rfloor$ and use $X = \log[D(A_y)/D(A_y^*)]$, including the same rounded reference for odd sizes. Joint zero-intercept fits to $\Delta S = (c/3)X$ and $\Delta F = (k/\pi^2)X$ give

$$c = 1.0000072, \quad k = 0.9989488. \quad (10)$$

The weighted RMS residuals are 4.18×10^{-7} and 9.56×10^{-6} , respectively. These are deterministic fit discrepancies, not standard errors.

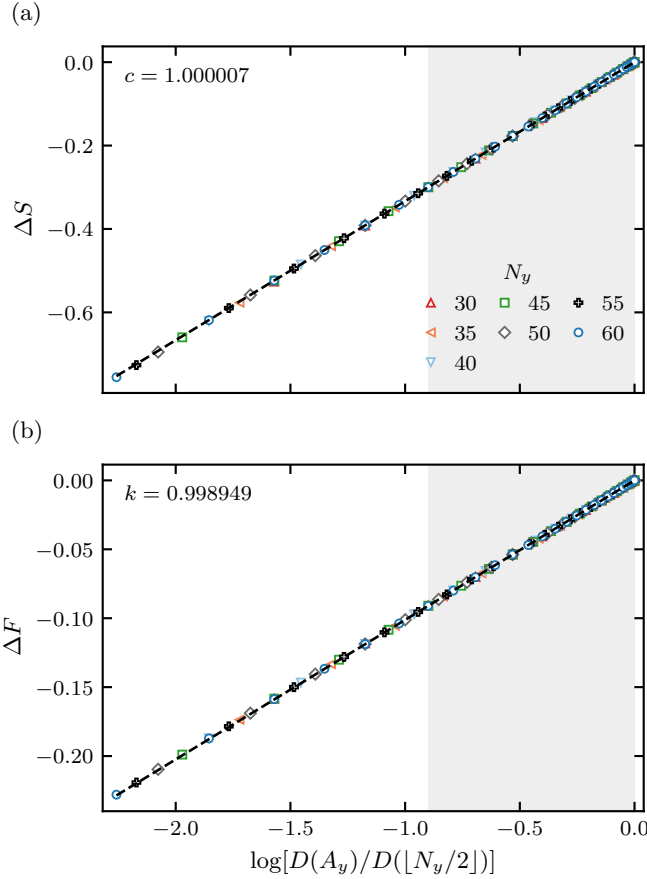


FIG. 5. Counterpart of manuscript Figure 6. Half-strip-subtracted (a) entropy and (b) charge variance for $N_y = 30, 35, 40, 45, 50, 55, 60$, $N_x = 20$, and $\alpha_1 = 1$. Points begin at $A_y = 2$; fits use $8 \leq A_y \leq \lfloor N_y/2 \rfloor$ with equal total weight per size. Shading marks the union of those fit windows.

VII. RESTRICTED SPECTRA: MANUSCRIPT FIGURE 7

The full occupation histogram uses 100 equal bins on $[0, 1]$. The conditional entanglement-energy histogram uses

$$\varepsilon = \log \frac{1 - \nu}{\nu}, \quad W : |2\nu - 1| < 0.99, \quad (11)$$

and 101 equal bins on $[-\log 199, \log 199]$. Both densities integrate to one. At $N_y = 32$, every translated half-strip has 640 occupations. There are 30 retained levels per cut for $\alpha_1 = 1$ and 28 for $\alpha_1 = 3$. Pooling the 32 identical translated spectra changes counts but not normalized densities; it does not provide independent samples or smooth a discrete spectrum.

The raw count $N_{0.99}(A_y)$ is an integer staircase. For the clean topological state it is exactly 30 throughout the prescribed fit window $5 \leq A_y \leq 16$. The best affine log-chord fit is therefore constant. Its R^2 is undefined because the total variation is zero. This differs from the smoothly growing trajectory-averaged count and does not contradict logarithmic entropy: eigenvalues can move within the fixed window while its count stays constant.

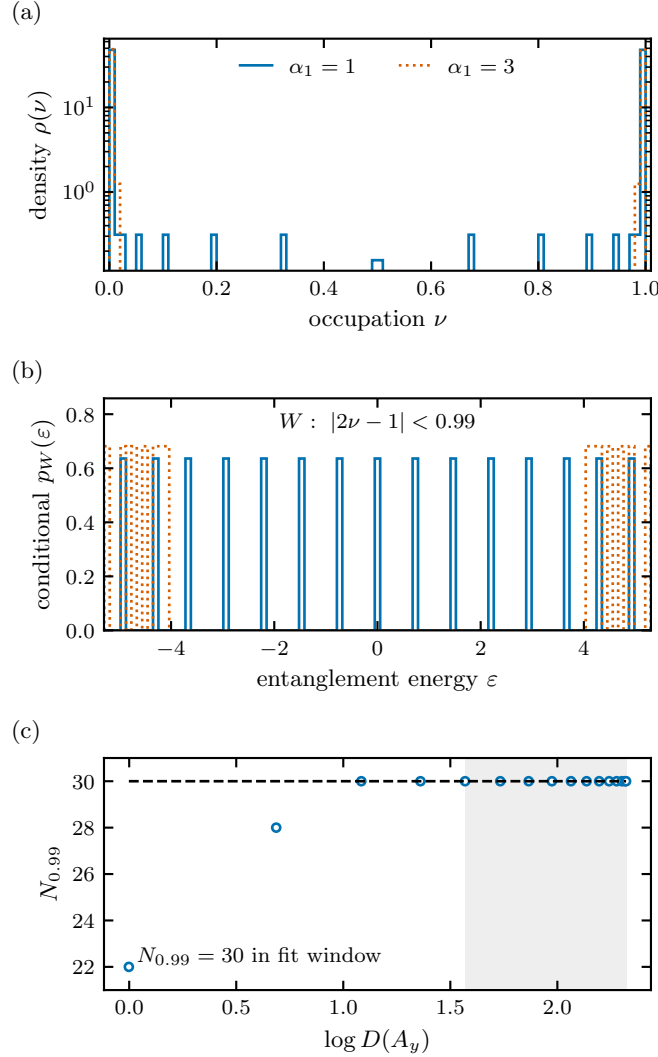


FIG. 6. Counterpart of manuscript Figure 7 at 20×32 . (a) Full occupation density, with zero-height bins omitted on the logarithmic axis. (b) Conditional entanglement-energy density within W . (c) Raw $\alpha_1 = 1$ window count versus log chord, with every strip width $1 \leq A_y \leq 16$ displayed. The shaded $5 \leq A_y \leq 16$ fit gives the constant 30. No ensemble broadening or statistical error bars are introduced.

VIII. ENTROPY CARRIED BY EACH WALL: MANUSCRIPT FIGURE 8

The additive Gaussian entropy contour of the full strip is

$$s(x, \delta y) = \sum_{\mu, a} |u_a(x, \delta y, \mu)|^2 h(\nu_a), \quad \sum_{x, \delta y} s(x, \delta y) = S(A_y). \quad (12)$$

We integrate over the two-column windows $x = 5, 6$ and $x = 14, 15$. These are contributions to the full-strip entropy, not entropies of separately traced narrow strips. With the same half-strip subtraction as above, the slopes are $m_L = 0.1666233$ and $m_R = 0.1666233$, giving

$$3m_L = 0.4998699, \quad 3m_R = 0.4998699. \quad (13)$$

Each wall contributes nearly one half of the full-strip coefficient. Equality of the two clean-parent values contrasts with the small asymmetry of the sampled trajectory curves. The leading coefficient carried outside these windows is small but need not vanish exactly. Entropy weights alone do not fix chirality.

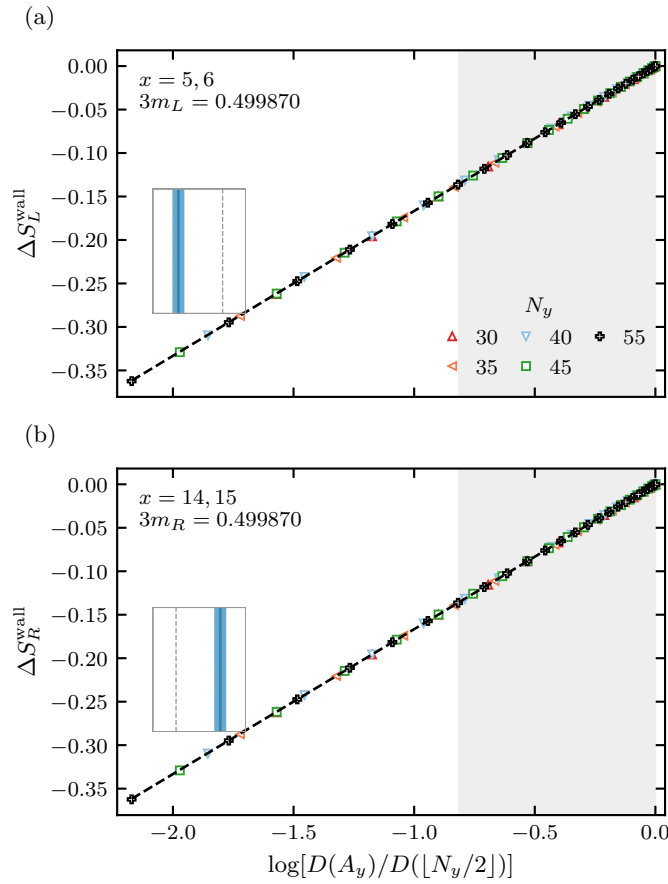


FIG. 7. Counterpart of manuscript Figure 8. Half-strip-subtracted contour sums for (a) the left window $x = 5, 6$ and (b) the right window $x = 14, 15$, at $N_y = 30, 35, 40, 45, 55$, $N_x = 20$, and $\alpha_1 = 1$. Insets mark the two integrated columns. Joint fits use $8 \leq A_y \leq \lfloor N_y/2 \rfloor$ and equal total weight per size; displayed widths begin at two.

IX. MODULAR EVOLUTION: MANUSCRIPT FIGURE 9

For a half-strip, use the manuscript correlation convention to define

$$h_A = \log[(\mathbf{1} - G_A)G_A^{-1}], \quad \varphi(t_{\text{mod}}) = e^{-ih_A^\top t_{\text{mod}}} \varphi(0). \quad (14)$$

Since $G_A = P_A^\top$, the calculation diagonalizes P_A and propagates with its entanglement energies. This explicitly preserves the transpose in the annihilation-profile convention. Centered occupations are clipped to $\pm(1 - \epsilon)$ with $\epsilon = 10^{-10}$ for the primary figure.

At each source $(x_s, \delta y) = (5, 8), (15, 8)$, the two orbital basis profiles are evolved separately and their densities added incoherently, with total norm two. The displacement uses the retained weight in the five-column source window:

$$\Delta y_{x_s}(t_{\text{mod}}) = \frac{\sum_{x=x_s-2}^{x_s+2} \sum_{\delta y=0}^{15} (\delta y - 8) p_{x_s}(x, \delta y; t_{\text{mod}})}{\sum_{x=x_s-2}^{x_s+2} \sum_{\delta y=0}^{15} p_{x_s}(x, \delta y; t_{\text{mod}})}. \quad (15)$$

The topological state's two displacements have opposite signs for all three cutoffs $10^{-8}, 10^{-10}, 10^{-12}$. Their magnitudes and oscillations are cutoff dependent: changing the cutoff shifts the curves by up to approximately 0.26 lattice cells over the displayed interval. The trivial displacement remains below 1.2×10^{-6} across these checks. Modular time is not circuit time; these curves diagnose handedness rather than a circuit transport velocity.

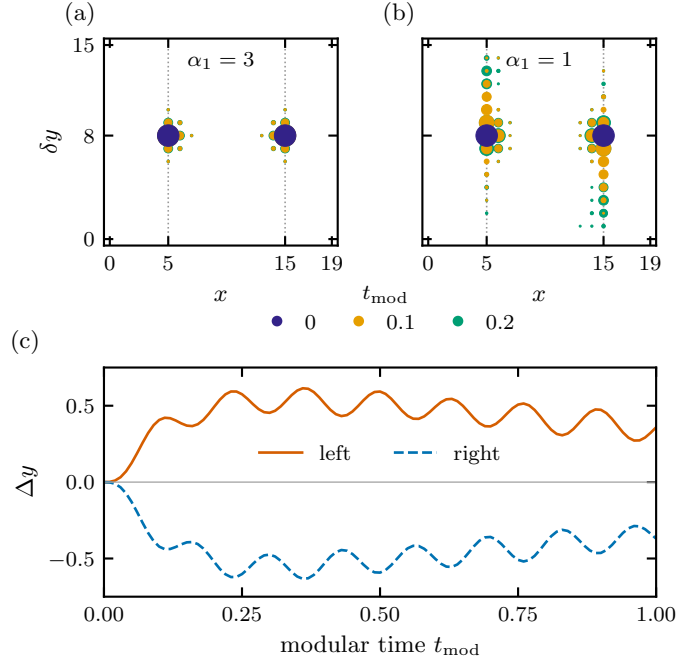


FIG. 8. Counterpart of manuscript Figure 9 at 20×32 , $A_y = 16$. Density snapshots at $t_{\text{mod}} = 0, 0.1, 0.2$ for (a) $\alpha_1 = 3$ and (b) $\alpha_1 = 1$. Marker areas follow the manuscript's density encoding. (c) Left- and right-wall displacements for $\alpha_1 = 1$, sampled every 0.01. The primary cutoff is 10^{-10} ; no SEM bands are applicable.

X. SPATIAL CONTOUR AND MUTUAL INFORMATION: MANUSCRIPT FIGURE A2

The topological contour contains weight along the walls in addition to the entanglement cuts. The trivial contour is concentrated near the cuts. For the mutual-information comparison, take opposite full-width strips a, b , each of width $\ell = N_y/4$, and compute

$$I_{a,b} = S(a) + S(b) - S(a \cup b). \quad (16)$$

Translation symmetry makes all distinct strip placements equivalent. The continuum two-interval reference retained from the manuscript is $I_{a,b} = (\log 2)/3$ for a unit free-fermion coefficient at this cross ratio. The deterministic parameter scan uses the exact nonuniform 21-point manuscript grid, including $\alpha_1 = 2$. At that point the zero Bloch vector is assigned $P_+(0) = P_-(0) = 1/2$ by the canonical construction; an open black ring identifies that explicitly prescribed point rather than interpolating through it silently.

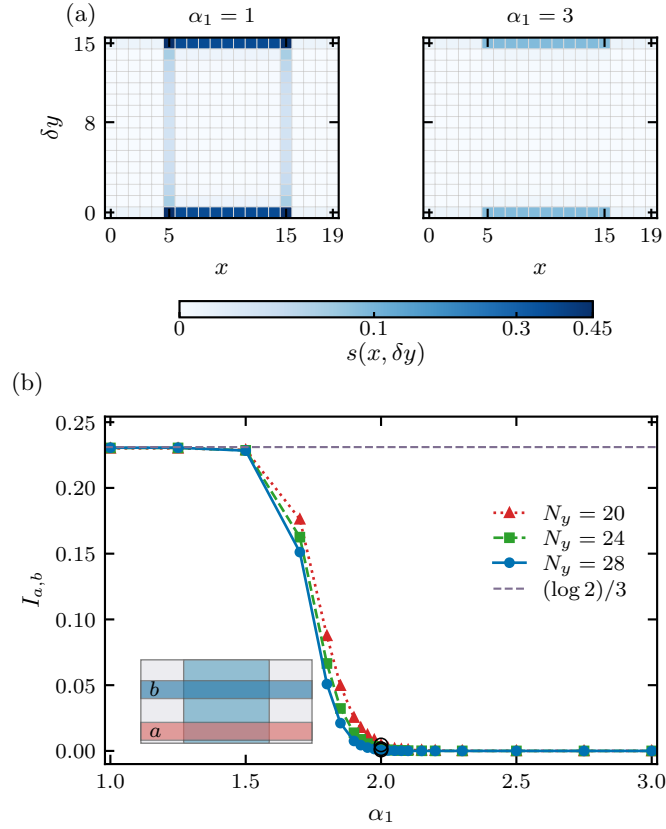


FIG. 9. Counterpart of manuscript Figure A2. (a) Ground-state half-strip entropy contours at 20×32 , $A_y = 16$, for $\alpha_1 = 1, 3$, with a shared square-root color scale. (b) Antipodal mutual information for $N_y = 20, 24, 28$ and $N_x = 20$. The dashed line is $(\log 2)/3$; lines connect sampled parameters. The inset identifies the two separated strips. All results are from the clean periodic parent at exact half filling.

XI. NUMERICAL VALIDATION AND INTERPRETATION

The result bundle contains 79 deterministic configurations, source and configuration hashes, all plotted arrays, and CSV tables. Direct dense assembly on a small lattice validates the momentum-block construction against the signed OW sum. Additional checks cover projector rank and idempotency, translation symmetry, hard-wall separation, Chern signs, the finite marker sum rule, Gaussian entropy and charge identities, normalized histograms, and modular norm conservation. A small exact Fock-space calculation independently verifies the factor of two and transpose in Eq. (6).

The earlier equilibrium density benchmark used a 10^{-7} shift of the momentum argument of representative OW columns before reconstruction in the untwisted periodic basis. Repeating that prescription on the same 20×30 input gives $g_{\text{hf}} = 1.41074 \times 10^{-7}$, compared with 1.50222×10^{-9} here. This is a protocol difference: the new benchmark retains the strictly periodic parent requested for this note. The earlier periodic 20×60 wall analysis is separately reproduced on its saved projector and correlator data; its cotangent fit explains why the present antipodal chord fit does not collapse.

The requested comparisons separate three conclusions. First, bulk topology, unit entropy and charge coefficients, equal half-unit wall entropy weights, and opposite modular handedness have close clean equilibrium counterparts. Second, the periodic lowest gap and the antipodal correlation shape depend on the finite-system boundary sector and the observable used. Third, the clean restricted spectrum is discrete, so a hard window count can plateau even while entropy varies logarithmically. These distinctions should remain visible if the benchmarks are later incorporated into an appendix.

A. Reproduction

Run the local CPU scripts from this directory:

```
python benchmark.py --threads 4
python analyze.py --threads 4
python validate.py --threads 4
python render.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=build parent_hamiltonian_benchmark_figs.tex
python verify.py --record
```

The final verifier copies the successfully built PDF to this directory and records the delivered files. Per-configuration NPZ/JSON pairs allow verified completed calculations to be reused. Invalid source identities or checksums trigger recomputation. Figure typography uses the existing manuscript style with this note's own inclusion widths and output directory. No circuit simulations, cloud jobs, or manuscript figure replacements are involved.